NORFOLK STATE UNIVERSITY

Department of Computer Science

SpartanGO

A Smart Campus Shuttle Application for Norfolk State University

CSC498 – Senior Capstone I (Fall)

Professional Project Proposal and CSC499 Implementation Baseline

Advisor: Dr. Nwosu
09/20/3036



Team Members	Tijaina Hill • Dennis Jordan • Jaden Lunsford • Ty'vaughn Daniels • HerMajesty Williamson
Faculty Advisor	Dr. Nwosu
CSC498 Focus	Analysis • Requirements • Feasibility • Design • Prototyping • Implementation Planning
CSC499 Focus	Implementation • Integration • Testing • Refinement • Deployment / Demonstration

Fall Semester → Spring Semester

CSC498 establishes an implementation-ready design baseline. CSC499 executes the approved design.

1. Executive Summary

SpartanGO is a proposed smart-campus shuttle application intended to improve the predictability, usability, and security of Norfolk State University shuttle transportation. The concept centers on three student-facing capabilities: requesting a ride from designated stops, receiving live arrival information and estimated wait times, and presenting a digital QR-based verification credential for boarding. The project also anticipates limited driver or administrative functions needed to manage requests, shuttle status, and verification workflows.

The project is intentionally organized as a two-semester senior capstone. CSC498 (Fall) is the analysis, requirements, feasibility, architecture, user-experience design, security planning, and implementation-planning semester. CSC499 (Spring) is the implementation, integration, testing, refinement, and final demonstration semester. This separation prevents premature coding before the team has validated data access, transportation workflows, university constraints, and the technical feasibility of location, notification, and identity-verification services.

Semester Boundary

No production application is required in CSC498. Any code written in CSC498 should be limited to disposable feasibility experiments when necessary to answer a technical question. The principal CSC498 outcome is an implementation-ready System Requirements and Design Baseline for CSC499.

2. Problem Statement and Project Rationale

Students who rely on campus shuttle transportation may experience uncertainty when they cannot see a shuttle's current status, estimate when it will arrive, or determine whether a ride request has been received. The preliminary proposal identifies this lack of real-time visibility and ride availability information as a central source of friction. A well-designed digital service could reduce unnecessary waiting, improve communication between riders and shuttle operations, and provide a more consistent boarding-verification workflow.

SpartanGO therefore proposes a unified mobile experience that presents shuttle-related information and actions in one place. However, the project must account for important feasibility constraints: access to reliable shuttle-location data, the availability of driver or transportation-office workflows, university identity and security policies, privacy implications of location data, and the practicality of using a QR-based credential. CSC498 will explicitly investigate and document these constraints before implementation begins.

3. Project Goal and Objectives

The overall project goal is to design and, in CSC499, implement a secure and usable prototype that demonstrates how a smart shuttle application could improve campus transportation for NSU students while respecting operational, privacy, and institutional constraints.

1. Document the current shuttle-use problem, target users, operating assumptions, and core user journeys.
2. Define functional and non-functional requirements for ride requests, shuttle status and ETA information, notifications, and QR-based verification.
3. Assess the technical and institutional feasibility of live shuttle location, mapping, notification, and identity-verification capabilities.
4. Design the student mobile experience and any minimum driver/administrator interface required to support the core workflows.
5. Design the system architecture, data model, interface contracts, security controls, and failure-handling behavior.
6. Produce measurable acceptance criteria and a comprehensive CSC499 test plan before production implementation begins.

7. Create an implementation backlog, responsibility matrix, and staged CSC499 development plan that the team can execute in Spring.

4. Two-Semester Project Scope

The following scope intentionally separates design work from implementation work. Scope changes after CSC498 should require advisor approval so that CSC499 remains focused and achievable.

Capability	CSC498 (Fall)	CSC499 (Spring)
Student ride request	Requirements, workflow, wireframes, validation rules, data model, acceptance criteria	Implement request submission, queue/status updates, cancellation rules, and user feedback
Live shuttle status / ETA	Investigate data source options; define refresh rate, map/stop model, ETA assumptions and fallback behavior	Implement approved location/ETA approach using real or simulated data as authorized
Notifications	Define events, message content, timing, permission model and failure behavior	Implement push or in-app notifications using the selected technology
QR verification	Define threat model, token lifecycle, privacy requirements, display/scan workflow, and university-policy dependencies	Implement approved demo verification workflow; do not represent it as an official NSU credential unless explicitly authorized
Driver / admin support	Define minimum roles, screens, request queue and validation workflow	Implement only the approved minimum operational interface
Security / privacy	Threat analysis, access-control design, data-minimization rules, logging specification	Implement controls and perform security/failure testing
Testing	Write test cases, success metrics and traceability matrix	Execute functional, integration, usability, performance and security tests

4.1 CSC498 In-Scope Activities

- Background and comparable-system research
- Stakeholder and workflow analysis
- Requirements elicitation and prioritization
- Feasibility analysis for mapping/GPS, notifications, identity/QR, hosting and data access
- High-fidelity wireframes or clickable UI prototype
- Logical and deployment architecture design
- Database/entity-relationship design and interface specifications
- Security, privacy, misuse and failure-mode analysis
- Testing strategy, acceptance criteria and requirements traceability
- CSC499 implementation backlog, schedule, ownership and risk plan

4.2 CSC498 Explicitly Out of Scope

- Production mobile-app implementation
- Production backend/API implementation
- Operational shuttle tracking deployment
- Official replacement of NSU student identification
- Live integration with university transportation, identity, or notification systems without approval

- Production hosting or campus-wide release
- Formal acceptance testing of a production system

5. Stakeholders and Primary Use Cases

Stakeholder	Interest / Responsibility
NSU Students	Request rides, view shuttle status, receive updates, and present a boarding-verification token.
Shuttle Drivers / Operators	View or respond to ride demand, communicate status, and validate boarding as appropriate.
Transportation / Campus Operations	Define operational rules, approved stops, service hours, and data availability.
University IT / Security	Review authentication, identity, data protection, hosting, integrations, and QR-verification approach.
Faculty Advisor / Evaluators	Assess capstone scope, engineering rigor, progress, documentation, and final demonstration.

The minimum demonstration should support coherent end-to-end scenarios rather than disconnected features. The CSC498 design package should therefore specify at least the following user journeys: student requests a ride from a designated stop; student checks shuttle/ETA status; student receives a status or arrival alert; student displays a time-limited verification code; authorized operator validates the code; and the system handles unavailable data or failed requests gracefully.

6. Preliminary Functional Requirements

ID	Requirement
FR-01	The system shall allow an authenticated student to identify or select an approved pickup stop.
FR-02	The system shall allow a student to submit a ride request and receive a clear request status.
FR-03	The system shall display available shuttle status, stop information, or estimated arrival information using the approved data source.
FR-04	The system shall provide notifications or in-app alerts for meaningful status changes defined during CSC498.
FR-05	The system shall generate and display a verification token or QR representation appropriate for the approved prototype workflow.
FR-06	The system shall allow an authorized operator interface to validate a prototype verification token.
FR-07	The system shall prevent expired, malformed, or otherwise invalid verification tokens from being treated as valid.
FR-08	The system shall provide understandable feedback when location, notification, identity, or network services are unavailable.
FR-09	The system shall maintain only the minimum data necessary to support approved project workflows.
FR-10	The system shall record selected operational or security events needed for testing and troubleshooting without exposing unnecessary personal data.

7. Non-Functional Requirements

Quality Attribute	CSC498 Design Expectation
Usability	Core student workflows should require few steps, provide visible status, and remain understandable without training.
Security	Authentication, authorization, token validation, secure transport, secrets handling, and least-privilege access must be designed before implementation.
Privacy	Location and identity-related data should be minimized, purpose-limited, and retained only as required for the prototype.
Reliability	The design must define behavior when GPS/location data, network connectivity, notifications, or backend services fail.
Performance	CSC498 will establish measurable response-time and refresh targets appropriate to the selected architecture.
Maintainability	Modules and interfaces should be separated so that mapping, notification, and identity components can be changed without redesigning the entire system.
Accessibility	UI design should use readable typography, meaningful labels, sufficient contrast, and accessible interaction patterns.
Auditability	Important actions such as ride requests and verification outcomes should be traceable during testing without excessive collection of personal data.

8. Proposed System Architecture

The preliminary architecture below separates the student and operator interfaces from shared application services. This separation supports modular development in CSC499 and makes institutional dependencies explicit. The actual technologies remain a CSC498 design decision and should be selected only after the team compares feasible alternatives.

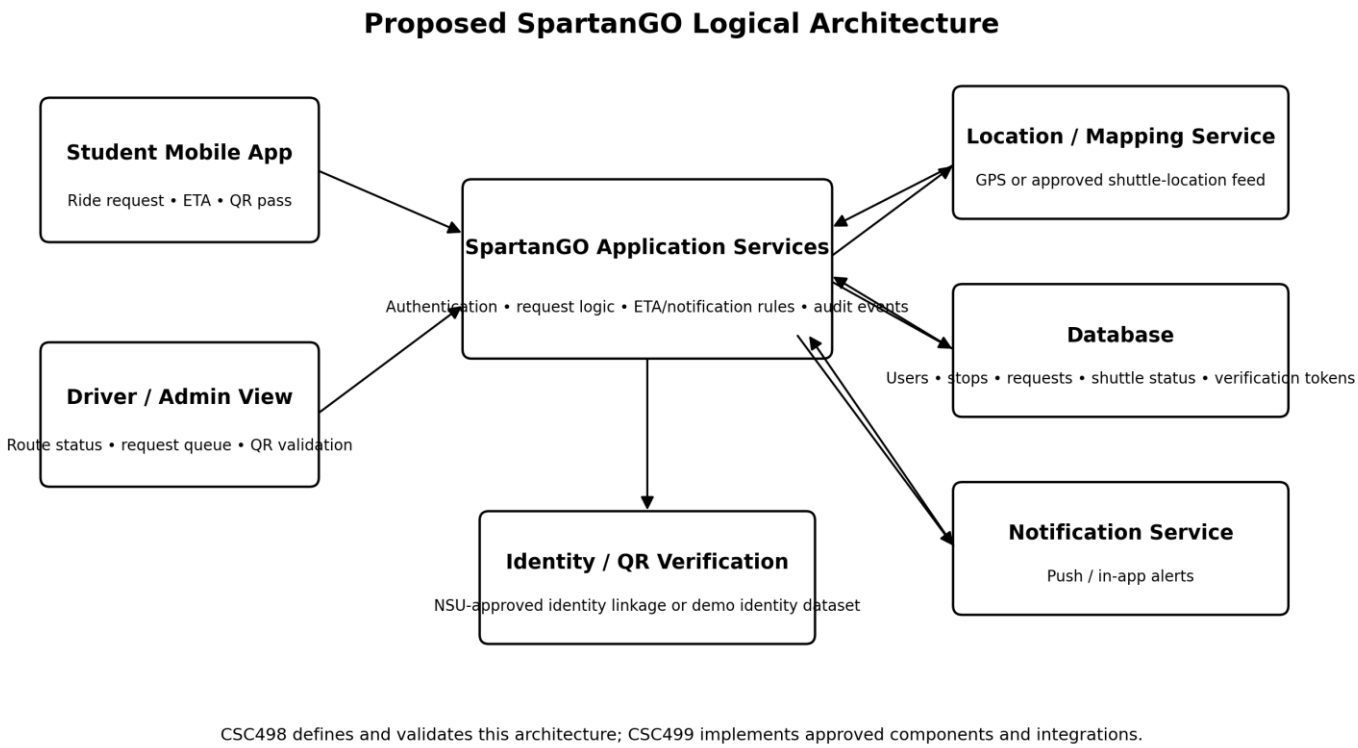


Figure 1. Proposed logical architecture for SpartanGO.

8.1 Architectural Components to Specify in CSC498

- Student mobile interface and navigation model
- Minimum driver/administrator interface
- Application service/API boundary
- Ride-request state model
- Stop, shuttle and location-data model
- Notification trigger and delivery design
- QR/token generation and validation design
- Authentication and authorization model
- Logging, monitoring and failure-handling design
- External-service abstraction for mapping/location and notifications

9. Data and Database Design

CSC498 should produce an entity-relationship model and data dictionary before any production database is built. A preliminary model may include the entities below, but the final schema should reflect validated requirements and data-minimization principles.

Entity	Representative Data / Purpose
User	Internal prototype identifier, role, authentication linkage, permitted profile fields
Shuttle Stop	Stop identifier, name, coordinates or map reference, active status

Entity	Representative Data / Purpose
Shuttle / Route	Prototype shuttle identifier, route or service status, current assignment
Ride Request	Requester, pickup stop, timestamp, status, assignment/reference
Location Update	Shuttle reference, time, location/stop reference; retention minimized
Notification	Trigger, recipient reference, delivery status, timestamp
Verification Token	Short-lived token metadata, issuance/expiration, validation status; avoid storing raw sensitive credential data
Audit Event	Selected security and workflow events needed for testing and troubleshooting

10. CSC498 Feasibility Studies and Design Decisions

Several project elements cannot be responsibly implemented until their feasibility is established. CSC498 should document the evidence and decision for each item below.

Decision Area	Questions CSC498 Must Resolve	Required Output
Shuttle location source	Is approved real-time location data available? If not, what simulated or operator-entered alternative will be used?	Selected source, update model, accuracy assumptions, fallback plan
Mapping / ETA	Will ETA be stop-based, distance-based, route-based, or simulated? What accuracy can be claimed?	Defined algorithm/assumption and measurable evaluation method
Notifications	Which delivery mechanism is realistic for the prototype? What permissions and reliability limits exist?	Selected mechanism and event matrix
Student identity / QR	What may be represented in a prototype? What must not be portrayed as an official NSU ID?	Approved demo identity model and QR threat/validation design
Authentication	What accounts or demo identities will be used? What roles are necessary?	Authentication/authorization specification
Mobile framework	Which framework best fits team skills, device targets, maps, QR and notifications?	Technology comparison and justified selection
Backend / hosting	What backend and hosting approach supports the demo securely and affordably?	Deployment architecture and environment plan

11. Security, Privacy, and Ethical Considerations

SpartanGO combines location-related data, identity/verification, and transportation activity, so the design must avoid treating security as a final-semester add-on. CSC498 should produce a compact threat model and explicitly define what the prototype stores, who may access it, how long information is retained, and which actions require authorization.

- Use least privilege for student, driver/operator, and administrator roles.
- Do not embed secrets, API keys, or privileged credentials in a mobile client.
- Use encrypted network transport for all application-service communication.
- Design QR tokens to be short-lived or otherwise resistant to simple reuse; avoid encoding sensitive personal data directly in the QR payload.
- Minimize storage of precise location history unless it is demonstrably necessary for the prototype.

- Separate demonstration identity data from any official university credential unless formal authorization is obtained.
- Define safe failure behavior when location or identity services are unavailable.
- Include abuse cases such as repeated fake ride requests, token replay, unauthorized scanning, account misuse, and excessive request generation in the test plan.

12. User Experience and Prototype Design

CSC498 should produce a high-fidelity clickable prototype that demonstrates the complete student journey and the minimum operator workflow without requiring production code. The prototype should be used to validate navigation, terminology, status visibility, error handling, and the number of interactions needed to complete key tasks.

Required CSC498 Screen / Flow	Design Intent
Sign-in / entry	Clarify account context and role.
Student home / shuttle status	Show current service state and the next useful action.
Select pickup stop	Make designated stops easy to identify.
Ride request confirmation	Prevent accidental requests and show resulting status.
Live/estimated arrival view	Present time/location confidence clearly; distinguish estimates from facts.
Notification/status history	Explain changes without forcing repeated manual checking.
QR verification display	Provide an obvious, time-bounded verification state.
Operator validation view	Show valid / invalid / expired result clearly.
Service unavailable / degraded state	Explain what the user can do when data or connectivity fails.

13. Project Methodology and Team Process

The team should use an iterative engineering process during CSC498, but iteration in Fall is focused on improving requirements, prototypes, architecture, and specifications rather than building the production application. Weekly advisor reviews should be used to resolve open decisions, validate artifacts, and control scope. All major requirements and design changes should be reflected in a maintained project repository or shared workspace.

- Maintain a prioritized requirements backlog with acceptance criteria.
- Assign an owner and reviewer to each major artifact.
- Use version control for design documents, prototypes where practical, test specifications, and CSC499 code once implementation begins.
- Record architecture decisions and rejected alternatives so later changes have context.
- End CSC498 with a frozen baseline or advisor-approved change-control process for CSC499.

14. CSC498 Fall Semester Work Plan

Week	Focus	Primary Deliverable
1	Project confirmation and advisor alignment	Refine problem statement, team working agreement, project repository/workspace
2	Current-state research and stakeholder analysis	Campus shuttle workflow assumptions, comparable-system notes, stakeholder map
3	Requirements elicitation	Draft functional/non-functional requirements and user stories
4	Scope and acceptance criteria	MVP boundary, prioritized backlog, measurable acceptance criteria
5	UX information architecture	Navigation map, low-fidelity wireframes, key user journeys
6	Technology feasibility I	Mobile framework, backend, database and hosting comparison
7	Technology feasibility II	Location/mapping/ETA, notification and QR/identity feasibility findings
8	Architecture design	Logical architecture, module boundaries, interfaces and external dependencies
9	Data and security design	ERD/data dictionary, roles/permissions, threat model, privacy/retention rules
10	High-fidelity prototype	Clickable student workflow and minimum operator/validation workflow
11	Interface and failure specifications	API/interface contracts, state transitions, degraded-service behavior
12	Testing design	Requirements traceability, test cases, security/failure tests, usability plan
13	CSC499 implementation planning	Sprint/backlog plan, module ownership, environment setup plan, risk controls
14	Design review and revision	Advisor design review; resolve gaps; finalize implementation-ready baseline
15	CSC498 final presentation / handoff	Present design baseline, prototype, feasibility results, Spring implementation plan

15. Required CSC498 Deliverables

Deliverable	Expected Content
D1. Final CSC498 Proposal	Approved project definition, two-semester scope, objectives and constraints.
D2. Background and Feasibility Report	Comparable systems and technical/institutional feasibility decisions.
D3. Software Requirements Specification	Functional/non-functional requirements, user stories, acceptance criteria and traceability.
D4. UI/UX Design Package	User journeys, wireframes and high-fidelity clickable prototype.
D5. System Architecture Package	Architecture diagram, component responsibilities, interfaces and deployment plan.
D6. Data Design Package	ERD/schema specification, data dictionary and retention/minimization decisions.
D7. Security and Privacy Plan	Roles, threat model, QR/token design, misuse cases and safeguards.
D8. Testing and Evaluation Plan	Test cases, metrics, usability approach, failure tests and success criteria.

Deliverable	Expected Content
D9. CSC499 Implementation Plan	Prioritized backlog, module ownership, sprint/milestone schedule and environment plan.
D10. CSC498 Final Presentation	Professional design review and Spring implementation readiness demonstration.

16. Team Organization and Accountability

The five team members should finalize individual assignments with the advisor after CSC498 feasibility work clarifies the selected technology stack. To preserve shared ownership, each major area should have a primary owner and at least one reviewer. Suggested workstreams are listed below; assigning them does not prevent cross-team collaboration.

Workstream	CSC498 Responsibility	CSC499 Responsibility
Product / Requirements	Requirements, backlog, traceability, advisor coordination	Change control, acceptance verification, demo coordination
Mobile UX / Front End	Wireframes, interaction design, navigation specification	Mobile interface implementation and UI tests
Backend / Data	API design, ERD, data rules, backend technology selection	Service, database and request-state implementation
Location / Notifications	Feasibility, ETA/location model, event matrix	Approved mapping/location and notification implementation
Security / QR / Testing	Threat model, token design, test specification	Verification workflow, security/failure testing, integration quality

17. Key Risks and Mitigation Strategy

Risk	Impact	Mitigation
No access to live NSU shuttle telemetry	High	Design a simulated or operator-updated location source in CSC498 so CSC499 can proceed without misrepresenting data as production telemetry.
Official student-ID integration is not permitted	High	Use a clearly labeled prototype identity and short-lived QR verification token; do not claim official-ID replacement.
ETA accuracy is poor	Medium	Define the ETA method and confidence/limitations; prefer stop/status estimates if real routing data is unavailable.
Push notification setup is unreliable or time-consuming	Medium	Select a feasible notification approach in Fall and retain in-app status as a fallback.
Scope expands to full transportation management	High	Freeze the approved MVP at the end of CSC498; defer routing optimization, payments, fleet management and other extras.
Uneven team contribution	Medium	Use workstream ownership, weekly evidence of progress, peer review and advisor checkpoints.
Sensitive location/identity data is over-collected	High	Apply data minimization, short retention, role-based access and demo data wherever production data is unnecessary.

18. CSC498 Success Criteria

CSC498 is successful when the team has removed enough uncertainty that CSC499 can begin implementation immediately. The Fall semester should therefore be evaluated on design readiness rather than on the amount of application code produced.

- All core requirements are documented, prioritized, testable, and traceable to user needs.
- The ride-request, arrival/status, notification, and QR-verification workflows are fully represented in the UI prototype and system design.
- The team has an advisor-approved plan for location/ETA data, including a realistic fallback if live university data is unavailable.
- The QR/identity design clearly distinguishes a capstone prototype from any official NSU credential unless formal approval exists.
- The architecture, interfaces, database design, security rules, and failure behavior are specified sufficiently for implementation.
- The test plan contains measurable functional, usability, security, failure, and performance criteria.
- The CSC499 backlog identifies implementation order, ownership, dependencies, and milestones.
- There are no unresolved high-impact feasibility questions that would require re-scoping the project after Spring coding begins.

19. CSC499 Spring Implementation Plan

CSC499 should execute the CSC498 baseline rather than restarting requirements and design. Minor refinements are expected, but major scope or architecture changes should be justified and approved because they reduce time available for integration and testing.

CSC499 Stage	Implementation Focus	Expected Evidence
Stage 1 – Environment and Skeleton	Repositories, environments, authentication shell, database migrations, core app navigation	Buildable baseline and development environment
Stage 2 – Core Ride Workflow	Student stop selection, ride request, request state and operator queue	End-to-end ride-request workflow
Stage 3 – Shuttle Status / ETA	Approved map/location/status approach and ETA presentation	Working location/status view with documented limitations
Stage 4 – Notifications	Selected push/in-app status notifications	Verified trigger and delivery behavior
Stage 5 – QR Verification	Short-lived token generation/display and operator validation	Valid/invalid/expired verification tests
Stage 6 – Integration	Unify mobile, backend, data, location, notification and verification components	Integrated prototype
Stage 7 – Test and Refine	Functional, integration, usability, security, failure and performance testing	Test report, defect log and corrections
Stage 8 – Finalize	Documentation, deployment/demo packaging, presentation and final evaluation	Final prototype and capstone presentation

20. Expected Final Project Outcome

At the end of CSC499, the team should be able to demonstrate a coherent SpartanGO prototype in which a student can interact with designated shuttle stops, submit and track a ride request, view approved shuttle-status or ETA information, receive meaningful status notifications, and present a prototype QR verification token that can be validated by an authorized operator interface. The demonstration should clearly distinguish implemented capabilities from simulated data sources and from any functionality that would require institutional deployment approval.

The final capstone value is not merely the set of screens. It is the engineering evidence that the team can analyze a real campus problem, design a secure and usable solution under practical constraints, implement the approved design, evaluate its behavior, and communicate both its capabilities and limitations professionally.

21. CSC498 → CSC499 Readiness Checklist

Readiness Item	Readiness Item
☐ Advisor-approved problem statement and MVP scope	☐ Complete requirements and acceptance criteria
☐ Approved technology stack and environment plan	☐ Validated approach for shuttle location / ETA data
☐ Approved prototype identity and QR-verification model	☐ High-fidelity clickable UI prototype
☐ Architecture and interface specifications	☐ ERD / data dictionary
☐ Security, privacy and threat model	☐ Test plan and requirements traceability matrix
☐ CSC499 backlog, milestones and team ownership	☐ Known risks and fallback plans documented

22. Advisor Review and Approval

Approval of this proposal authorizes the team to proceed with CSC498 analysis, requirements, feasibility, design, and implementation planning. It does not authorize production deployment, use of official NSU identity credentials, or access to operational university systems beyond what is separately approved by the appropriate university personnel.

Faculty Advisor: Dr. Nwosu

Advisor Signature: 

Date: 09/20/2026

Team Representative: HerMajesty Williamson

Date: 09/20/2026